Mukesh Patel School of Technology
Management & Engineering

QueryCart

An English query interface for a small inventory

Natural Language Processing (702AI0C009)

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1. Problem statement

A small-shop operator may need stock and price answers without writing SQL or spreadsheet formulas. A bounded language interface can make common inventory queries easier to express and inspect.

2. Objectives

- Recognize supported list, filter, rank, count and aggregation questions.
- Show the interpreted intent, slots and equivalent SQL preview.
- Execute a read-only local query and export the result.

3. Proposed approach

Symbolic semantic parsing uses complete grammar patterns and typed slots. The parsed plan is executed by JavaScript over a local CSV. The SQL preview is educational; no external database or trained text-to-SQL model is used.

4. Data and evaluation design

The demonstration inventory contains 24 fictional products and 288 physical units. Thirty-two original diagnostic questions include 20 supported questions and 12 unsupported requests.

Evaluation checks exact results for supported fixtures and correct rejection for unsupported fixtures. It measures the documented grammar on authored cases, not accuracy on arbitrary natural-language questions.

5. Methodology

Stage	Method	Purpose
Normalize	Unicode, case and whitespace	Match consistent surface forms
Parse	Whole-query grammar; typed slots	Identify intent and parameters
Validate	Category, ranges and CSV schema	Reject ambiguous requests
Execute	Read-only local table operation	Compute exact results
Explain	Plan, SQL preview and export	Make the interpretation visible

Language understanding, normalization, grammar matching, intent and slot extraction, semantic representation and question answering over structured data.

6. Application example

Input: Show Electronics under 1000.
Plan: category = Electronics and price < 1000.
Input: Total inventory value by category.
Plan: group by category and sum stock multiplied by unit price.

7. Results

Diagnostic	Correct / total
Supported exact results	20 / 20
Unsupported-query rejections	12 / 12
All authored fixtures	32 / 32

These 32 authored checks validate the intended grammar. They do not establish accuracy for arbitrary phrasing.

Interpretation

The supported questions return the expected inventory results, and the unsupported requests are rejected. The main limitation is phrasing coverage: a person may understand a question that does not match the current grammar. The visible query plan helps the user check what was understood.

User workflow

Enter text or choose an example, inspect the suggested fields or result, review the interpretation, then export the desired output. Computation runs in the browser; no paid inference API is required.

8. Limitations

- Only documented English grammar is supported.
- Questions such as Anything running out? can be meaningful but unsupported.
- No write queries, forecasts, sales history, arbitrary joins or currency conversion.
- The SQL preview is not a receipt of database execution.

9. Conclusion and future work

A typed query plan connects a small natural-language grammar to useful inventory operations. Exact rejection behavior is valuable because an apparently plausible table can otherwise hide a misunderstood request.

References

1. Jurafsky and Martin, Speech and Language Processing. https://web.stanford.edu/~jurafsky/slp3/

2. SQLite SELECT documentation. https://www.sqlite.org/lang_select.html

3. Live demo: querycart-dhruv.vercel.app. https://querycart-dhruv.vercel.app

4. Source code: github.com/dhruvpaleja/querycart-nlp. https://github.com/dhruvpaleja/querycart-nlp